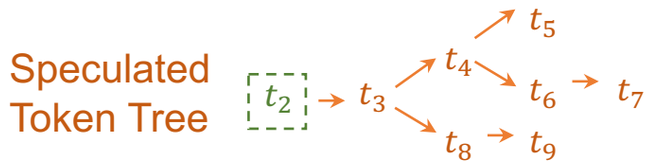

t_i Verified tokens

Available KV-cache slots

Verified tokens' KV-cache

Speculated tokens' KV-cache

Sequence-based Parallel Decoding

Kernel 1: $t_2 \rightarrow t_3 \rightarrow t_4 \rightarrow t_5$ Kernel 2: $t_2 \rightarrow t_3 \rightarrow t_8 \rightarrow t_9$ Kernel 3: $t_2 \rightarrow t_3 \rightarrow t_4 \rightarrow t_6 \rightarrow t_7$

KV-cache

t_2	t_3	t_4	t_5	…

t_2	t_3	t_8	t_9	…

t_2	t_3	t_4	t_6	t_7	…

Causal Mask

	t_2	t_3	t_4	t_5
t_2	✓	✓	✓	✓
t_3		✓	✓	✓
t_4			✓	✓
t_5				✓

	t_2	t_3	t_8	t_9
t_2	✓	✓	✓	✓
t_3		✓	✓	✓
t_8			✓	✓
t_8				✓

	t_2	t_3	t_4	t_6	t_7
t_2	✓	✓	✓	✓	✓
t_3		✓	✓	✓	✓
t_4			✓	✓	✓
t_6				✓	✓
t_7					✓

Tree-based Parallel Decoding

Kernel 1: speculated token tree

KV-cache

t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	…

Topology-Aware Causal Mask

	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9
t_2	✓	✓	✓	✓	✓	✓	✓	✓
t_3		✓	✓	✓	✓	✓	✓	✓
t_4			✓	✓	✓	✓		
t_5				✓				
t_6					✓	✓		
t_7						✓		
t_8							✓	✓
t_9								✓